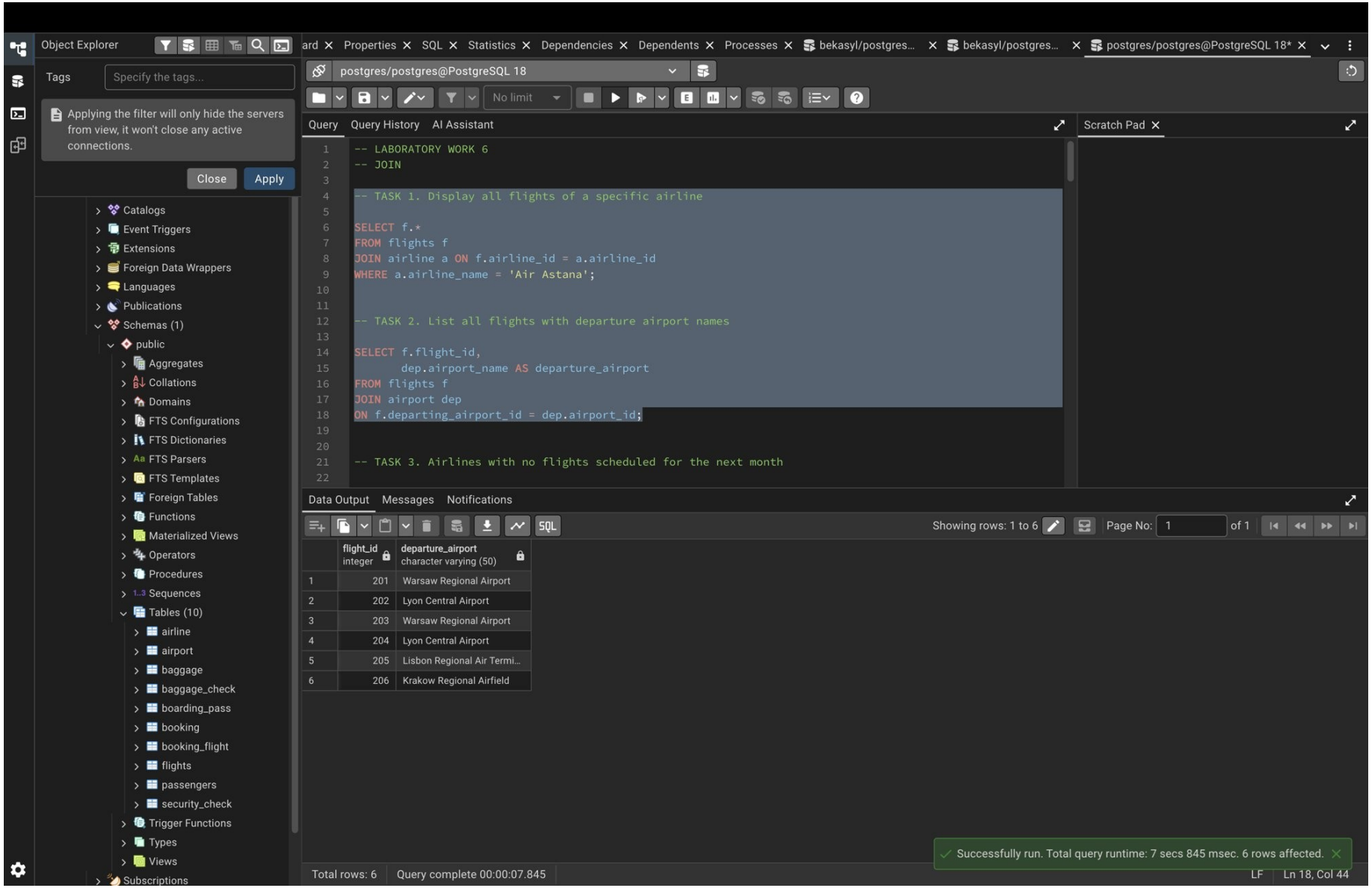
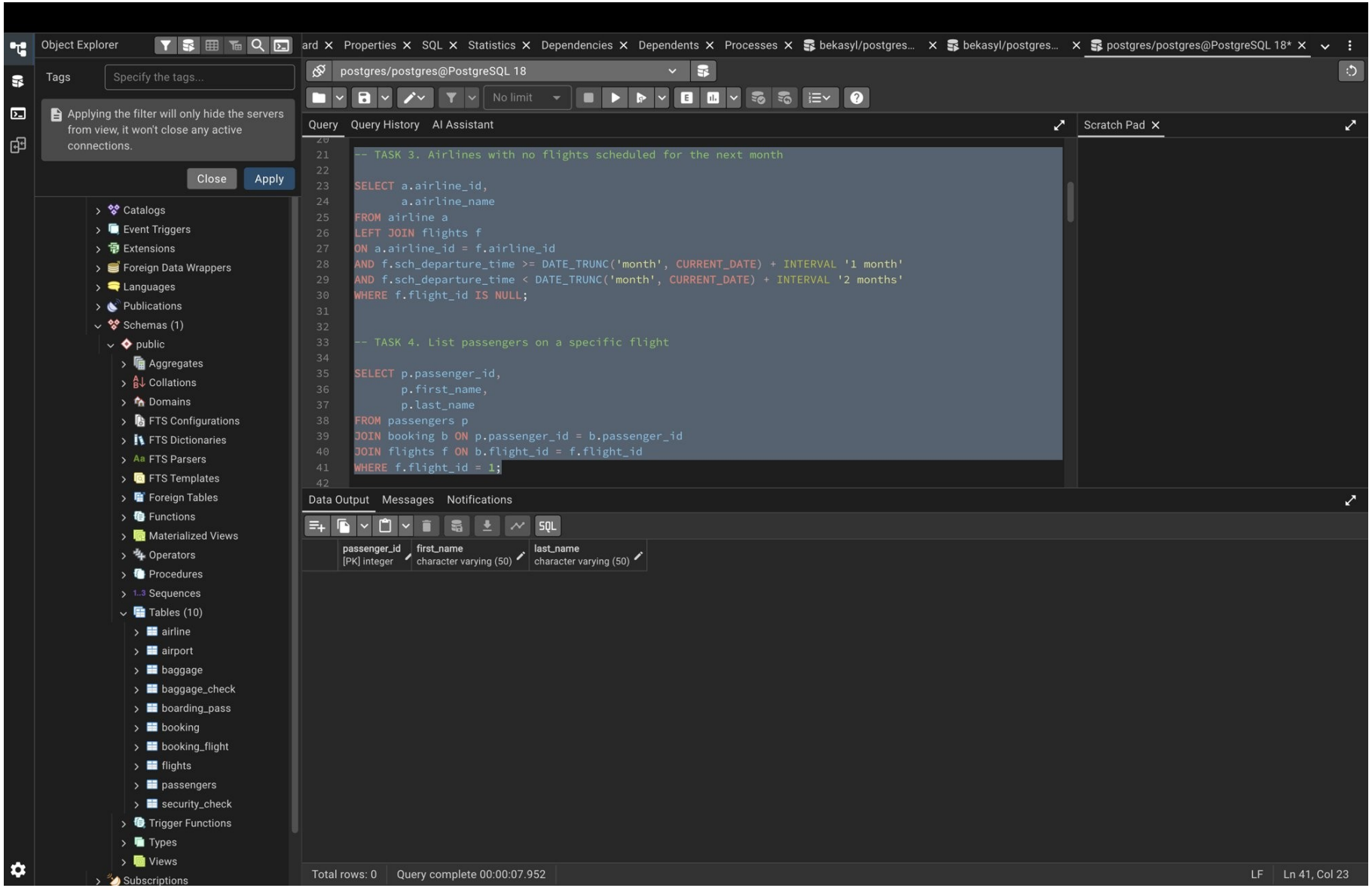
Laboratory Work 6

JOIN

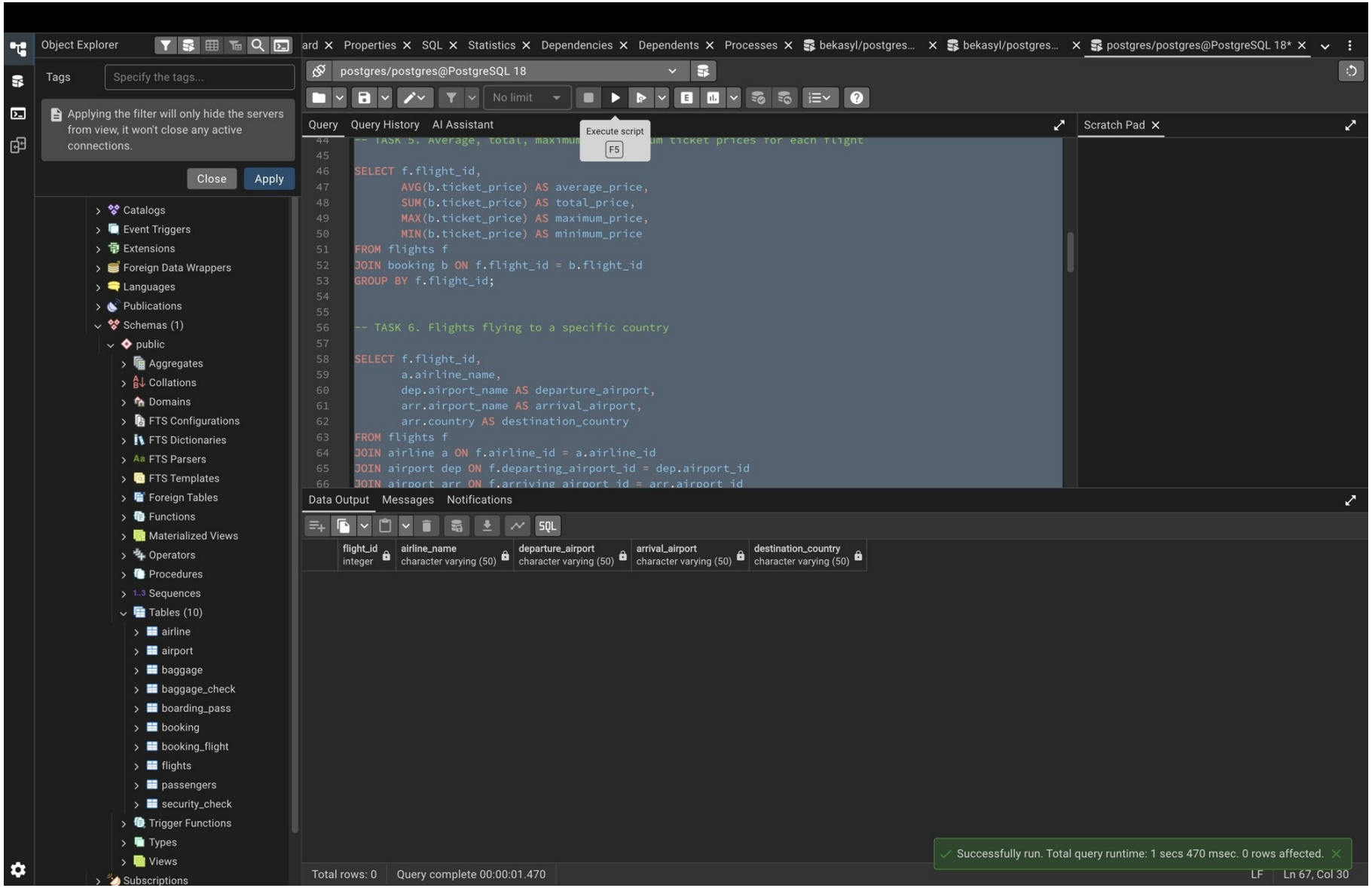
Task 1–2


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Applying the filter will only hide the servers from view, it won't close any active connections.
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bekasyl/postgres...
bekasyl/postgres...
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
-- LABORATORY WORK 6
-- JOIN
-- TASK 1. Display all flights of a specific airline
SELECT f.*
FROM flights f
JOIN airline a ON f.airline_id = a.airline_id
WHERE a.airline_name = 'Air Astana';
-- TASK 2. List all flights with departure airport names
SELECT f.flight_id,
dep.airport_name AS departure_airport
FROM flights f
JOIN airport dep
ON f.departing_airport_id = dep.airport_id;
-- TASK 3. Airlines with no flights scheduled for the next month
Data Output
Messages
Notifications
Showing rows: 1 to 6
Page No: 1 of 1
flight_id integer
departure_airport character varying (50)
1 201 Warsaw Regional Airport
2 202 Lyon Central Airport
3 203 Warsaw Regional Airport
4 204 Lyon Central Airport
5 205 Lisbon Regional Air Termi...
6 206 Krakow Regional Airfield
Successfully run. Total query runtime: 7 secs 845 msec. 6 rows affected.
Total rows: 6
Query complete 00:00:07.845
LF
Ln 18, Col 44

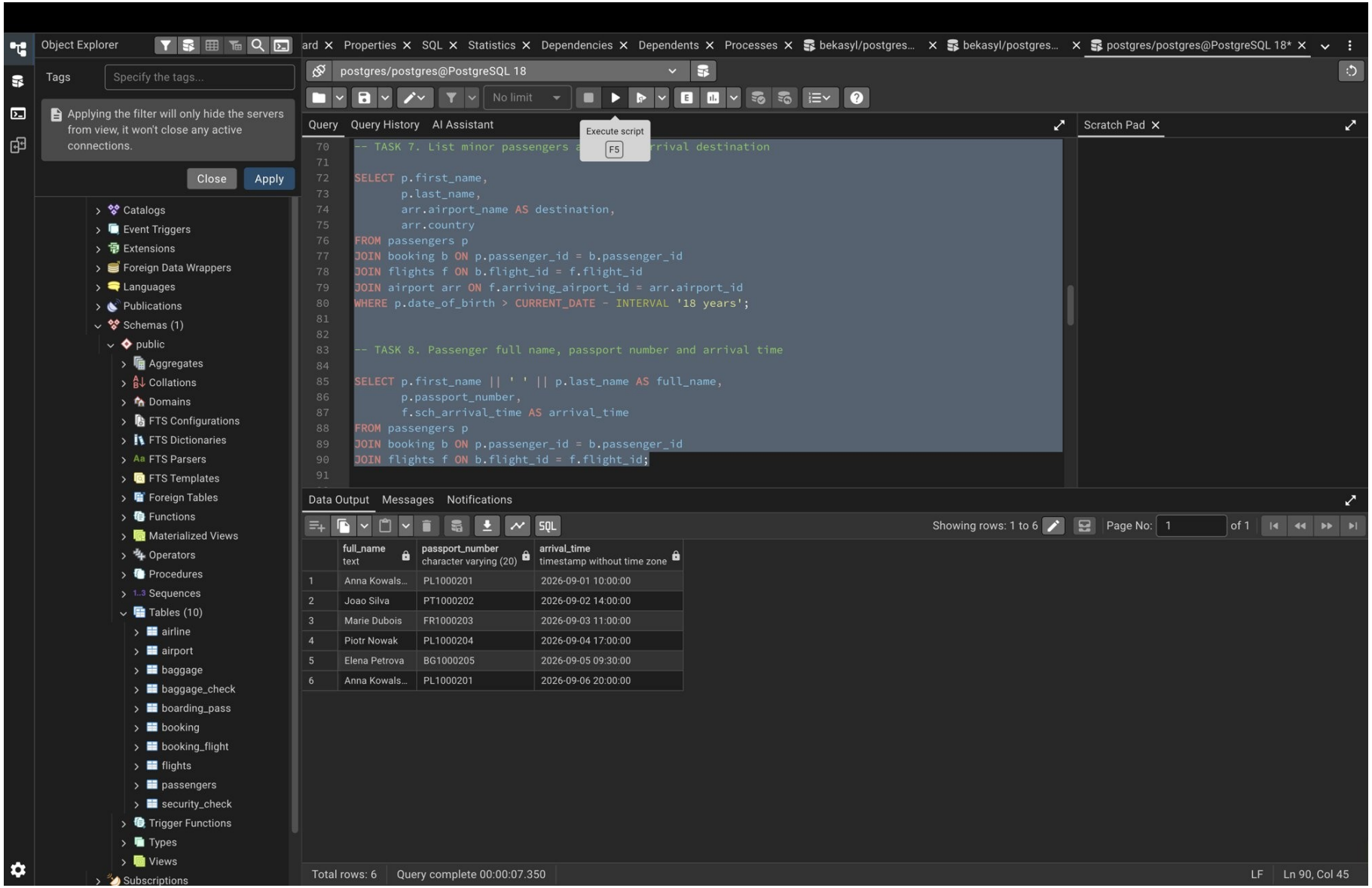
Task 3–4


Object Explorer
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Specify the tags...
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postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
-- TASK 3. Airlines with no flights scheduled for the next month
SELECT a.airline_id,
a.airline_name
FROM airline a
LEFT JOIN flights f
ON a.airline_id = f.airline_id
AND f.sch_departure_time >= DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '1 month'
AND f.sch_departure_time < DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '2 months'
WHERE f.flight_id IS NULL;
-- TASK 4. List passengers on a specific flight
SELECT p.passenger_id,
p.first_name,
p.last_name
FROM passengers p
JOIN booking b ON p.passenger_id = b.passenger_id
JOIN flights f ON b.flight_id = f.flight_id
WHERE f.flight_id = 1;
Data Output
Messages
Notifications
SQL
passenger_id
[PK] integer
first_name
character varying (50)
last_name
character varying (50)
Total rows: 0
Query complete 00:00:07.952
LF
Ln 41, Col 23

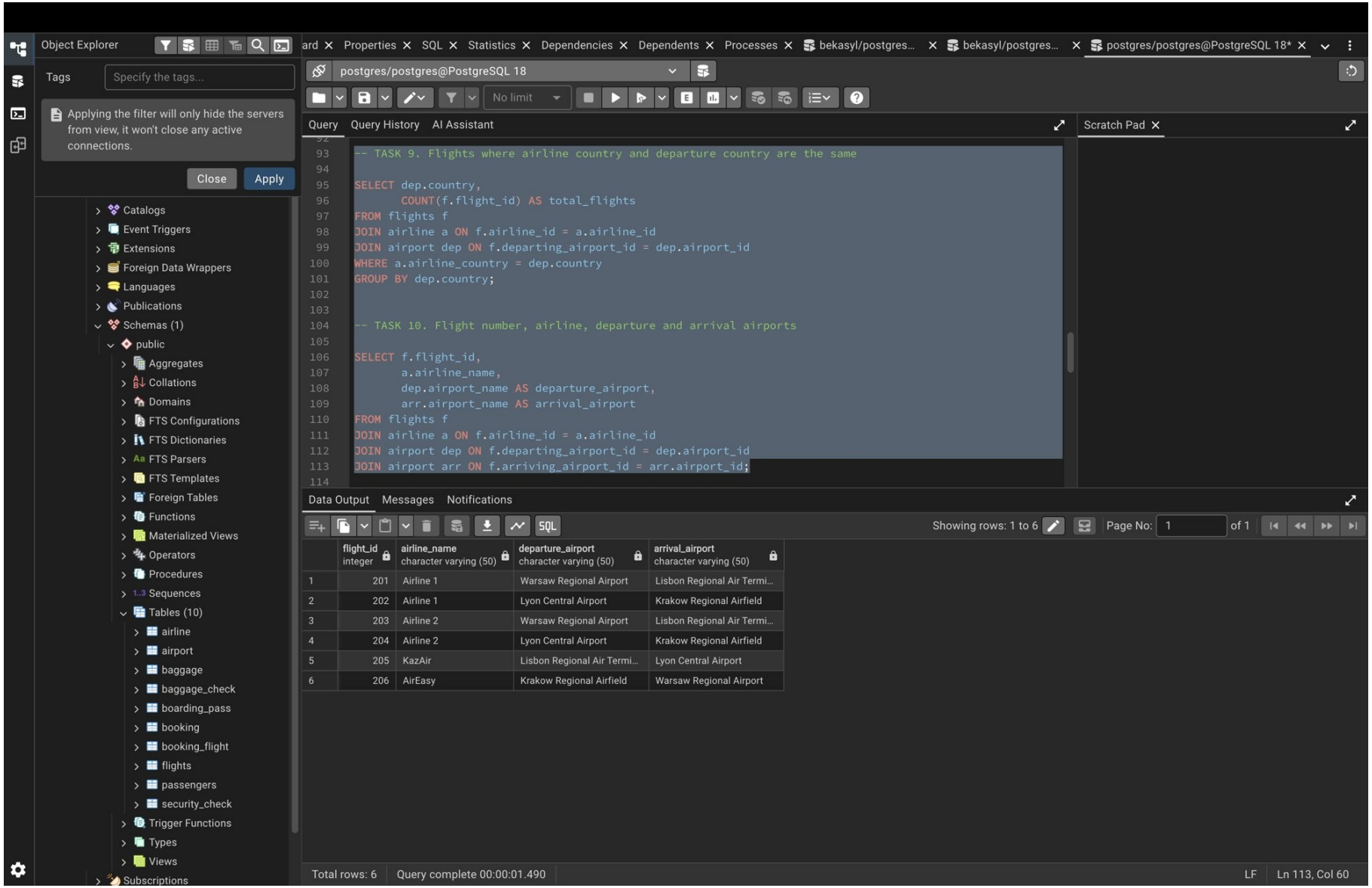
Task 5–6


Object Explorer
Tags
Specify the tags...
Applying the filter will only hide the servers from view, it won't close any active connections.
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No limit
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F5
Scratch Pad
-- TASK 5. Average, total, maximum ticket prices for each flight
SELECT f.flight_id,
AVG(b.ticket_price) AS average_price,
SUM(b.ticket_price) AS total_price,
MAX(b.ticket_price) AS maximum_price,
MIN(b.ticket_price) AS minimum_price
FROM flights f
JOIN booking b ON f.flight_id = b.flight_id
GROUP BY f.flight_id;
-- TASK 6. Flights flying to a specific country
SELECT f.flight_id,
a.airline_name,
dep.airport_name AS departure_airport,
arr.airport_name AS arrival_airport,
arr.country AS destination_country
FROM flights f
JOIN airline a ON f.airline_id = a.airline_id
JOIN airport dep ON f.departing_airport_id = dep.airport_id
JOIN airport arr ON f.arriving_airport_id = arr.airport_id
Data Output
Messages
Notifications
SQL
flight_id integer
airline_name character varying (50)
departure_airport character varying (50)
arrival_airport character varying (50)
destination_country character varying (50)
Successfully run. Total query runtime: 1 secs 470 msec. 0 rows affected.
Total rows: 0
Query complete 00:00:01.470
LF
Ln 67, Col 30

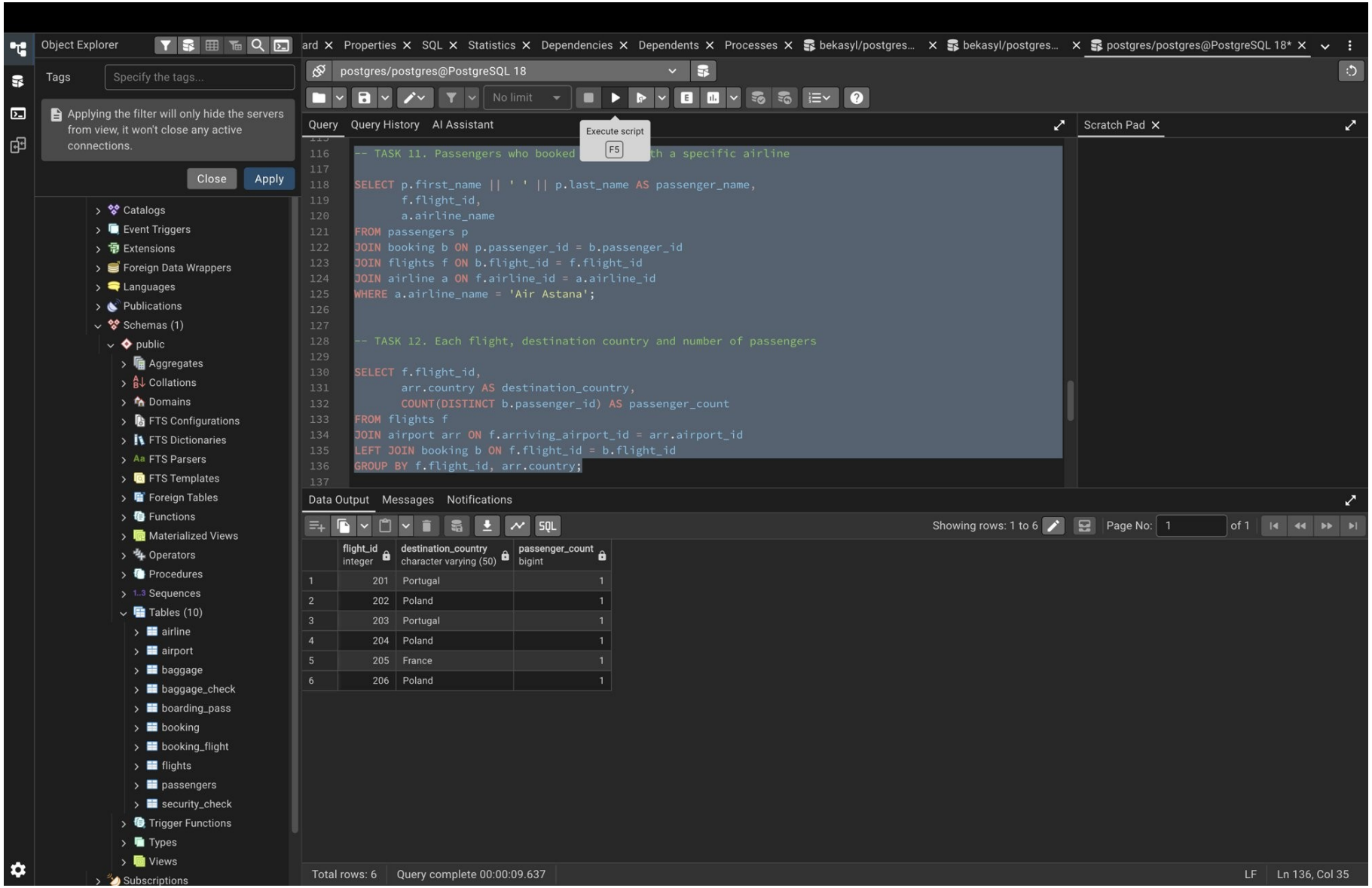
Task 7–8


Object Explorer
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Specify the tags...
Applying the filter will only hide the servers from view, it won't close any active connections.
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No limit
Query
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F5
Scratch Pad
-- TASK 7. List minor passengers
SELECT p.first_name,
p.last_name,
arr.airport_name AS destination,
arr.country
FROM passengers p
JOIN booking b ON p.passenger_id = b.passenger_id
JOIN flights f ON b.flight_id = f.flight_id
JOIN airport arr ON f.arriving_airport_id = arr.airport_id
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years';
-- TASK 8. Passenger full name, passport number and arrival time
SELECT p.first_name || ' ' || p.last_name AS full_name,
p.passport_number,
f.sch_arrival_time AS arrival_time
FROM passengers p
JOIN booking b ON p.passenger_id = b.passenger_id
JOIN flights f ON b.flight_id = f.flight_id;
Data Output
Messages
Notifications
Showing rows: 1 to 6
Page No: 1 of 1
full_name text
passport_number character varying (20)
arrival_time timestamp without time zone
1 Anna Kowals... PL1000201 2026-09-01 10:00:00
2 Joao Silva PT1000202 2026-09-02 14:00:00
3 Marie Dubois FR1000203 2026-09-03 11:00:00
4 Piotr Nowak PL1000204 2026-09-04 17:00:00
5 Elena Petrova BG1000205 2026-09-05 09:30:00
6 Anna Kowals... PL1000201 2026-09-06 20:00:00
Total rows: 6
Query complete 00:00:07.350
LF
Ln 90, Col 45

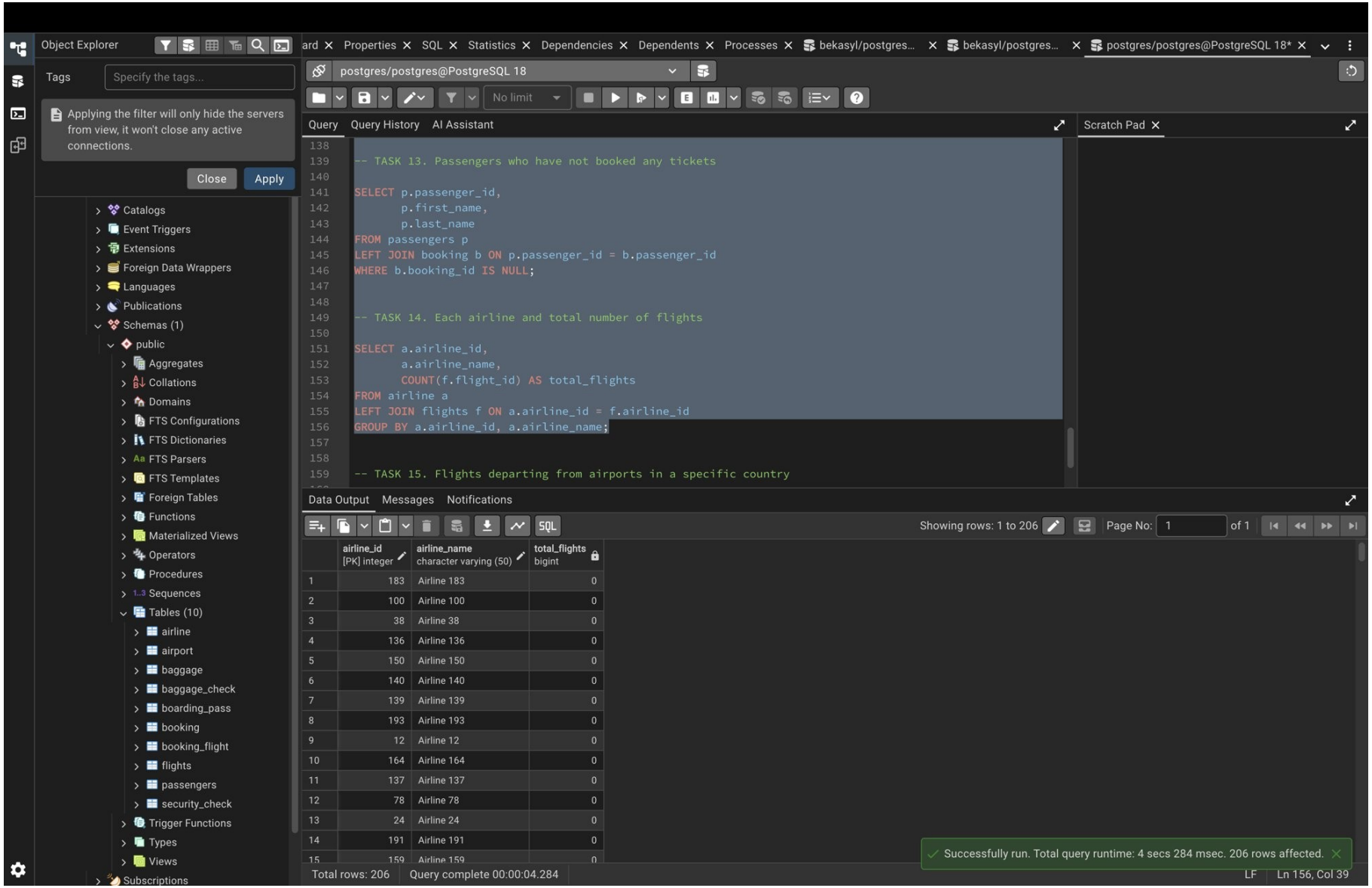
Task 9–10


Object Explorer
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Specify the tags...
Applying the filter will only hide the servers from view, it won't close any active connections.
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postgres/postgres@PostgreSQL 18
No limit
Query
Query History
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Scratch Pad
-- TASK 9. Flights where airline country and departure country are the same
SELECT dep.country,
COUNT(f.flight_id) AS total_flights
FROM flights f
JOIN airline a ON f.airline_id = a.airline_id
JOIN airport dep ON f.departing_airport_id = dep.airport_id
WHERE a.airline_country = dep.country
GROUP BY dep.country;
-- TASK 10. Flight number, airline, departure and arrival airports
SELECT f.flight_id,
a.airline_name,
dep.airport_name AS departure_airport,
arr.airport_name AS arrival_airport
FROM flights f
JOIN airline a ON f.airline_id = a.airline_id
JOIN airport dep ON f.departing_airport_id = dep.airport_id
JOIN airport arr ON f.arriving_airport_id = arr.airport_id;
Data Output
Messages
Notifications
Showing rows: 1 to 6
Page No: 1 of 1
flight_id integer
airline_name character varying (50)
departure_airport character varying (50)
arrival_airport character varying (50)
1 201 Airline 1 Warsaw Regional Airport Lisbon Regional Air Termi...
2 202 Airline 1 Lyon Central Airport Krakow Regional Airfield
3 203 Airline 2 Warsaw Regional Airport Lisbon Regional Air Termi...
4 204 Airline 2 Lyon Central Airport Krakow Regional Airfield
5 205 KazAir Lisbon Regional Air Termi... Lyon Central Airport
6 206 AirEasy Krakow Regional Airfield Warsaw Regional Airport
Total rows: 6
Query complete 00:00:01.490
LF
Ln 113, Col 60

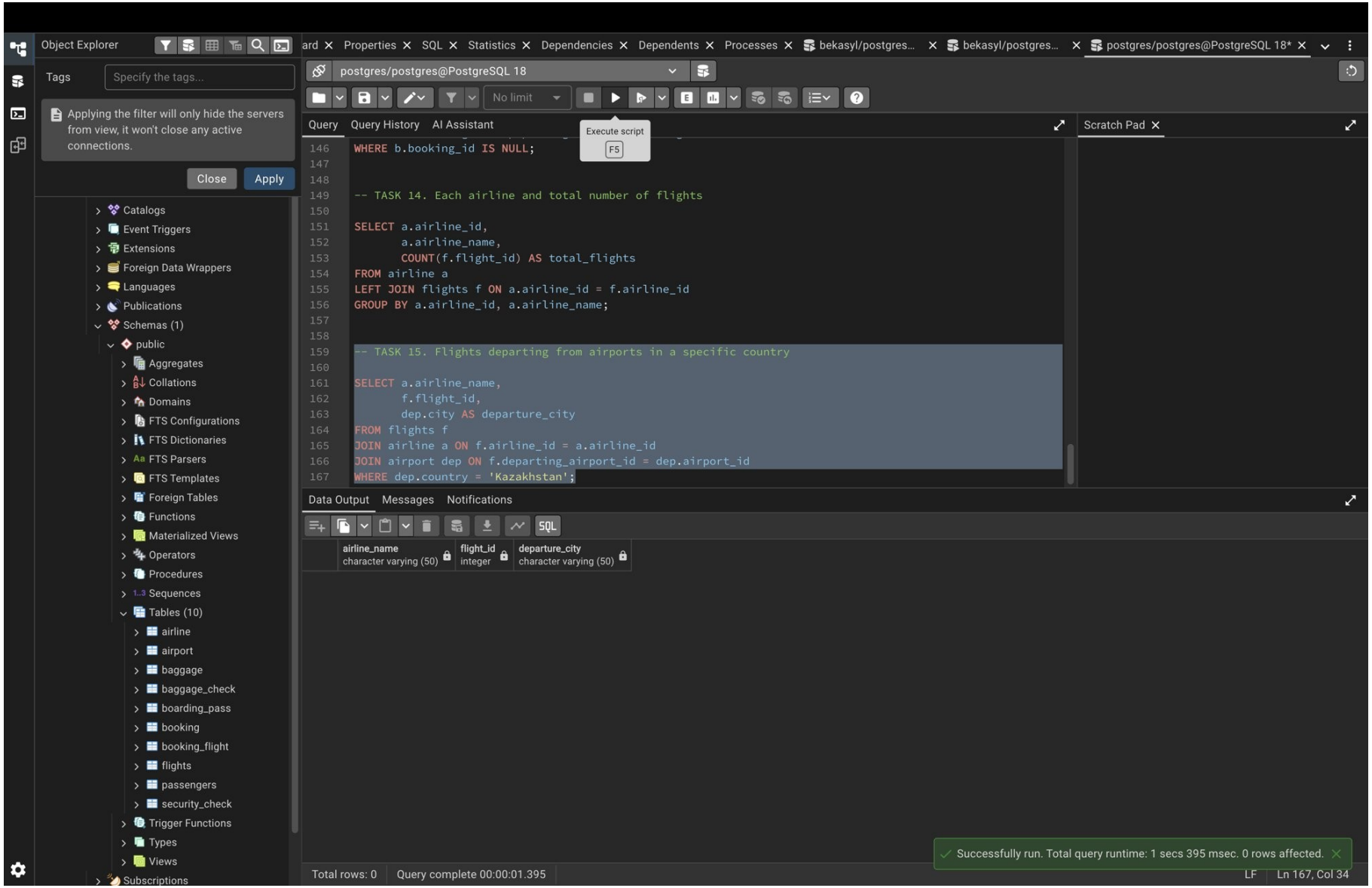
Task 11–12


Object Explorer
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Applying the filter will only hide the servers from view, it won't close any active connections.
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Scratch Pad
-- TASK 11. Passengers who booked
ch a specific airline
SELECT p.first_name || ' ' || p.last_name AS passenger_name,
f.flight_id,
a.airline_name
FROM passengers p
JOIN booking b ON p.passenger_id = b.passenger_id
JOIN flights f ON b.flight_id = f.flight_id
JOIN airline a ON f.airline_id = a.airline_id
WHERE a.airline_name = 'Air Astana';
-- TASK 12. Each flight, destination country and number of passengers
SELECT f.flight_id,
arr.country AS destination_country,
COUNT(DISTINCT b.passenger_id) AS passenger_count
FROM flights f
JOIN airport arr ON f.arriving_airport_id = arr.airport_id
LEFT JOIN booking b ON f.flight_id = b.flight_id
GROUP BY f.flight_id, arr.country;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1 of 1
flight_id integer
destination_country character varying (50)
passenger_count bigint
1 201 Portugal 1
2 202 Poland 1
3 203 Portugal 1
4 204 Poland 1
5 205 France 1
6 206 Poland 1
Total rows: 6
Query complete 00:00:09.637
LF
Ln 136, Col 35

Task 13–14


Object Explorer
Tags
Specify the tags...
Applying the filter will only hide the servers from view, it won't close any active connections.
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-- TASK 13. Passengers who have not booked any tickets
SELECT p.passenger_id,
p.first_name,
p.last_name
FROM passengers p
LEFT JOIN booking b ON p.passenger_id = b.passenger_id
WHERE b.booking_id IS NULL;
-- TASK 14. Each airline and total number of flights
SELECT a.airline_id,
a.airline_name,
COUNT(f.flight_id) AS total_flights
FROM airline a
LEFT JOIN flights f ON a.airline_id = f.airline_id
GROUP BY a.airline_id, a.airline_name;
-- TASK 15. Flights departing from airports in a specific country
Data Output
Messages
Notifications
Showing rows: 1 to 206
Page No: 1 of 1
airline_id [PK] integer
airline_name character varying (50)
total_flights bigint
1 183 Airline 183 0
2 100 Airline 100 0
3 38 Airline 38 0
4 136 Airline 136 0
5 150 Airline 150 0
6 140 Airline 140 0
7 139 Airline 139 0
8 193 Airline 193 0
9 12 Airline 12 0
10 164 Airline 164 0
11 137 Airline 137 0
12 78 Airline 78 0
13 24 Airline 24 0
14 191 Airline 191 0
Successfully run. Total query runtime: 4 secs 284 msec. 206 rows affected.
Total rows: 206
Query complete 00:00:04.284
LF
Ln 156, Col 39

Task 15


Object Explorer
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Specify the tags...
Applying the filter will only hide the servers from view, it won't close any active connections.
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WHERE b.booking_id IS NULL;
-- TASK 14. Each airline and total number of flights
SELECT a.airline_id,
a.airline_name,
COUNT(f.flight_id) AS total_flights
FROM airline a
LEFT JOIN flights f ON a.airline_id = f.airline_id
GROUP BY a.airline_id, a.airline_name;
-- TASK 15. Flights departing from airports in a specific country
SELECT a.airline_name,
f.flight_id,
dep.city AS departure_city
FROM flights f
JOIN airline a ON f.airline_id = a.airline_id
JOIN airport dep ON f.departing_airport_id = dep.airport_id
WHERE dep.country = 'Kazakhstan';
Data Output
Messages
Notifications
SQL
airline_name
character varying (50)
flight_id
integer
departure_city
character varying (50)
Successfully run. Total query runtime: 1 secs 395 msec. 0 rows affected.
Total rows: 0
Query complete 00:00:01.395
LF
Ln 167, Col 34